COPPERBELT UNIVERSITY

School of Medicine

Human Anatomy

MBS 200

End of Term Test one

27TH May, 2025

TIME: 9 to 12hrs *TOTAL MARKS: 153*

Student No.

INSTRUCTIONS FOR THE WHOLE PAPER

1. Write your computer number in the space above
2. Duration is 3 hours
3. Read the questions and instructions carefully before attempting to answer
4. Do not start writing until you are told to do so.

CELLPHONES & PROGRAMMABLE CALCULATORS ARE NOT ALLOWED IN THE EXAMINATION ROOM

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO BY THE INVIGILATOR

SECTION A: ANSWER ALL THE QUESTIONS IN THE SPACES PROVIDED

Q1 a. What is notochord? [2]

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b. Give its functions and fate. [4]

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Q2 a. What is capacitation of sperms? [3]

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b. How much time is required for it and what are the changes noted during capacitation? [4]

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Q3a. Describe the anatomical position [5]

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b. Explain why it is important to know the anatomical position [2]

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Q4. State the three embryonic gem layers and list three derivatives from each? [12]

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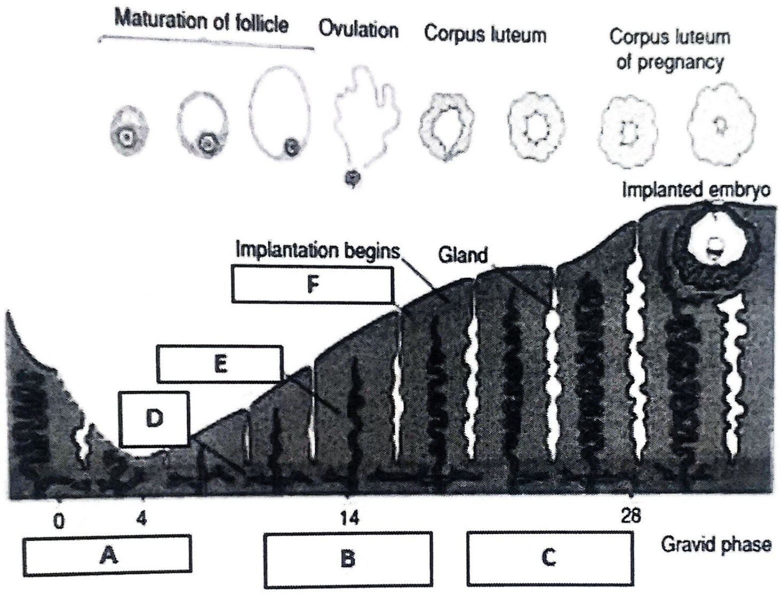
Q5. Normal Menstrual Cycle

The following diagram illustrates the menstrual cycle.



a. Label the phases of the menstrual cycle A, B and C.[3]

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b. The endometrium is the innermost layer of the uterus that sheds off during menses. It is depicted by layers D, E and F above. Label them and state their blood supply: [6]

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c. What are the other 2 layers of the uterus? [2]

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d. What type of glands are found in the uterus? [1]

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e. Which hormone is responsible for the following? [2]

i. Ovulation at Day 14

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ii.Phase B to occur

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f. Definitions:

i.Menorrhagia [2]

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ii.Dysmenorrhoea [1]

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....

iii.Metrorrhagia [2]

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Q6. The Placenta

Structurally, the placenta has two components: foetal and maternal portions.

a. Which chorion forms the foetal portion? [2]

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b. Which decidua forms the maternal portion? [1]

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c. State 4 differences between the foetal and the maternal portions of the placenta [4]

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d. List 4 protein hormones produced from the placenta and its function: [4]

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e. What type of placenta is the human placenta? [1]

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f. List the layers of the placental barrier from the maternal to the foetal side [6]

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Q7. A 24-year-old female with suspected meningitis experiences a delay in lumbar puncture due to poor sample labelling and processing. [4]

a. What stain is routinely used for CNS tissue histology?

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b. Which fibres primarily constitute the outer layer of meninges?

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c.What organelle is key for CSF-resorbing cells 'secretion?

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d. Which junctional complex prevents CSF leakage?

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Q8. An elderly diabetic patient presents with a chronic non-healing plantar ulcer at Ndola Teaching Hospital [4].

a. What type of epithelium covers the plantar skin?...

b. What receptor detects pressure changes?...

c. Which cytoskeletal protein anchors desmosomes? ....

e. What organelle would be abundant in phagocytic cells cleaning debris?

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Q9. A genetic profile was done to help diagnose a suspected chromosomal abnormality at a DNA Diagnostic centre. Chromosomal abnormalities maybe numerical or structural.

A. Define the following terms.

i. Aneuploid [1]

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ii. Trisomy and give two examples of a trisomy [3]

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iii. Monosomy and give two examples of a monosomy [3]

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iv. What can cause aneuploid? [1]

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v. what are structural chromosomal abnormalities? [2]

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vi. Mention two conditions that result from structural depletion of the chromosome 15, stating which one is on the maternal or paternal chromosome [4]

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Q10. 19-year-old footballer reports persistent back spasms after training. A muscle biopsy is performed to rule out myopathy [4].

a. What muscle type is biopsied?..

b. Which organelle stores calcium for contraction?..

c. Which muscle fibres resist fatigue?...

d. What neuromuscular structure initiates contraction?..

Q11a. **Draw and label the contents of full- term umbilical cord in the horizontal section [5]**

b. Name the two types of embryonic connective tissue [2]

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c. Describe the type of cells, fibers and state of ground substance found in the answers in (b)[6]

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d. The connective tissue of the head and neck is derived from which embryonic structures [1].

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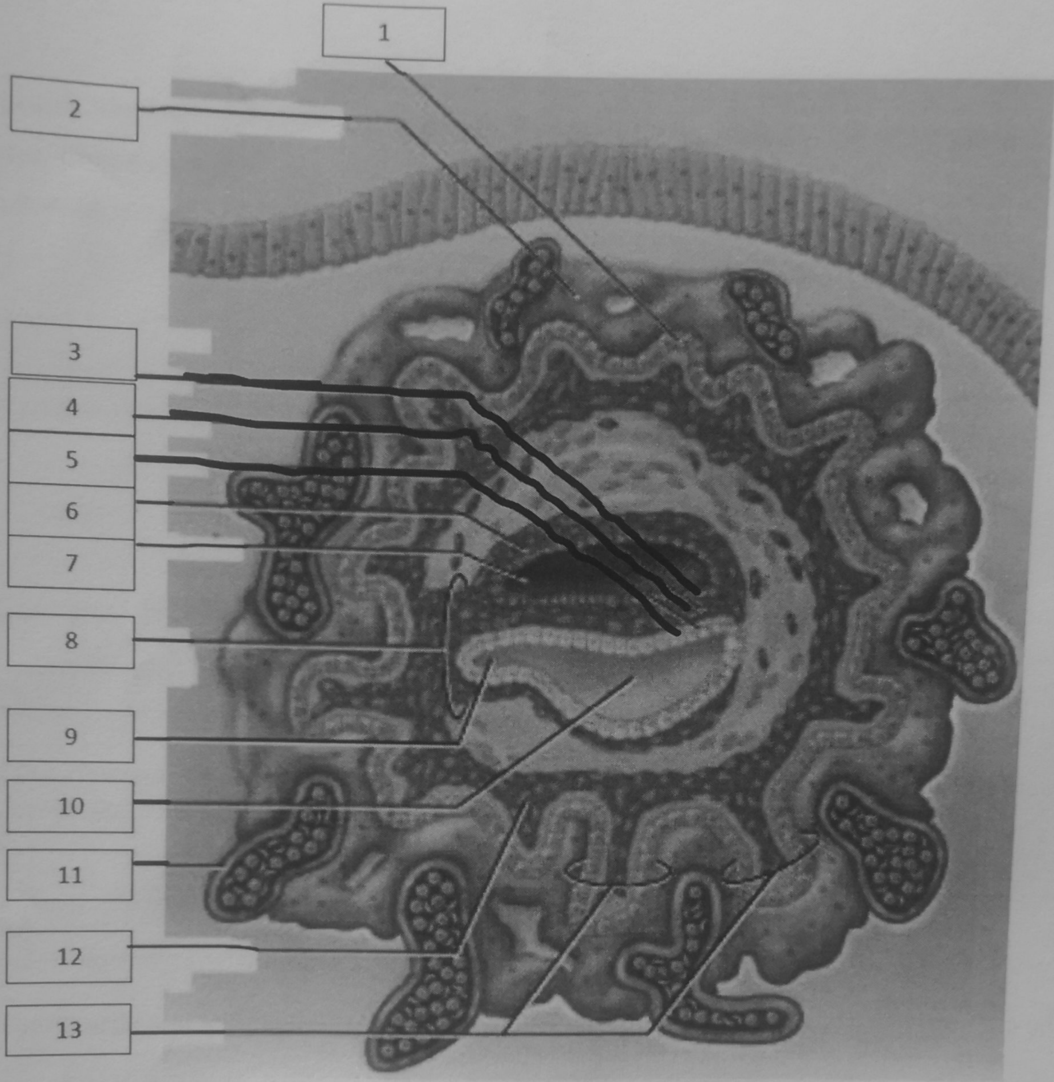
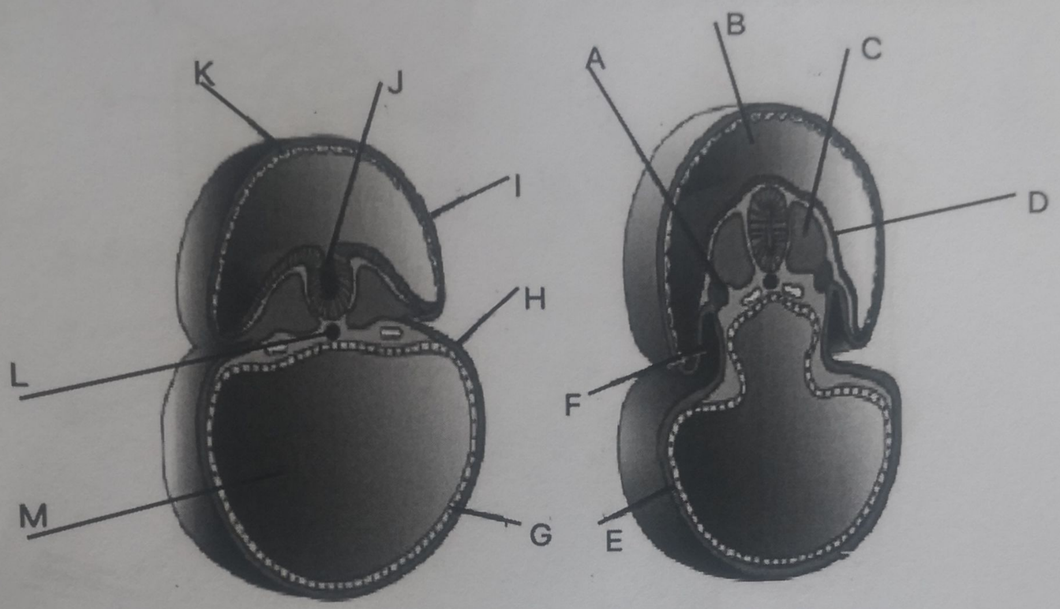
SECTION B: LABEL ALL THE DIAGRAMS

DIAGRAM A



1	8
2	9
3	10
4	11
5	12
6	13
7	

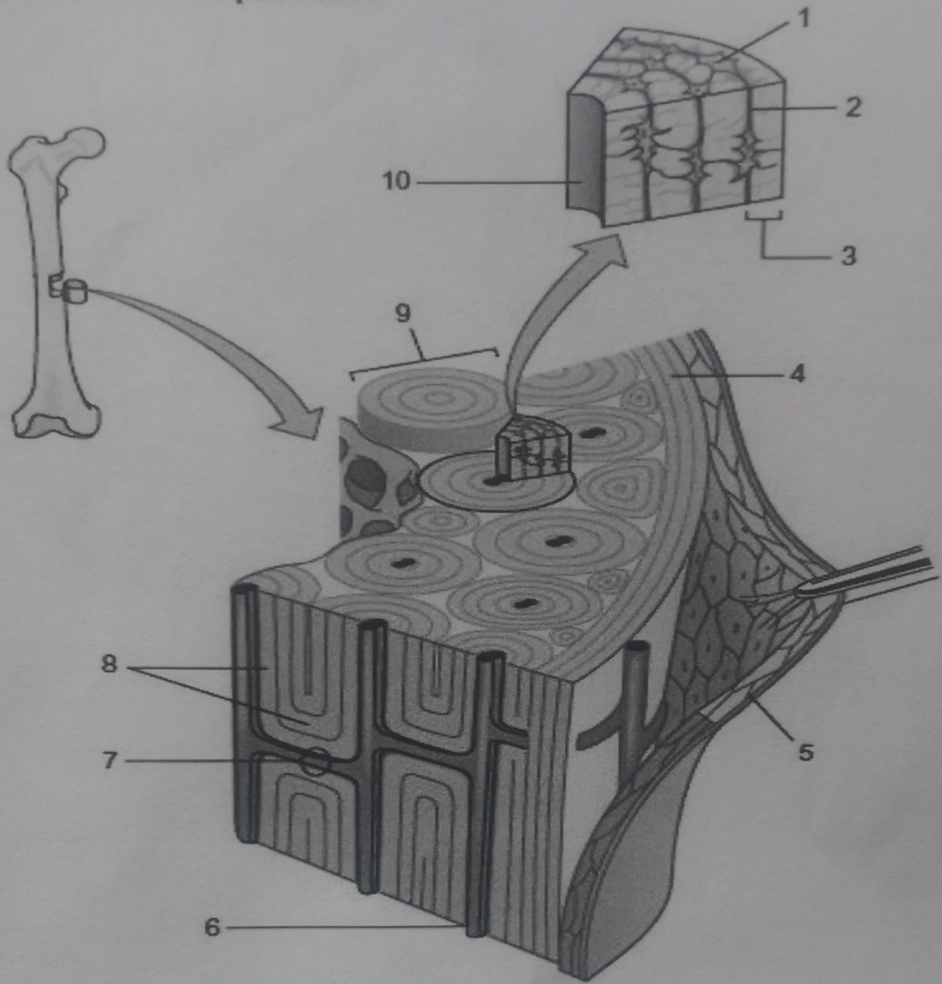
DIAGRAM B



A	H
B	I
C	J
D	K
E	L
F	M
G	

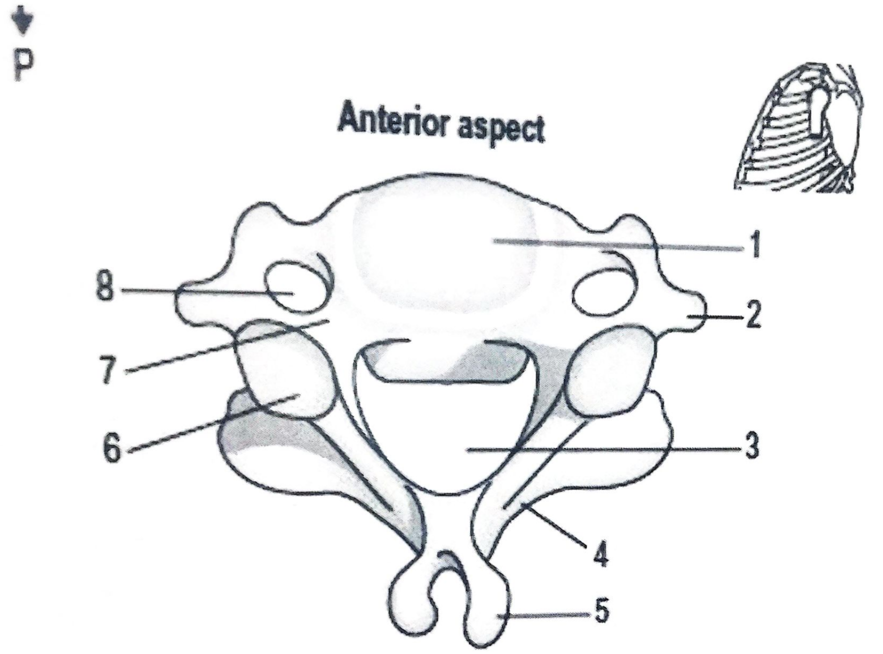
DIAGRAM C

Microstructure of compact bone



1	6
2	7
3	8
4	9
5	10

DIAGRAM D



1	5
2	6
3	7
4	8

TOTOZ

COLLECTION